DSP · L09

Digital filters

Cheat sheet · Day 17 · Mon 9 Nov · pack D10

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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L09 DIGITAL FILTERS
w0 = 2*pi*F0/Fs  (rad/sample);  F = w*Fs/(2*pi);  Omega = w*Fs
|H(w)| = |b0| * prod(dist to zeros)/prod(dist to poles); poles at 0 don't change |H|
LP: poles near z=1, zeros near z=-1.  HP: reverse.  BP: poles r e^{+-jw0}.
Linear phase: theta = -w*n0 ; tau = -d theta/dw ; symmetric FIR -> n0=(N-1)/2
  phase may jump by +-pi where real amplitude changes sign (still linear phase)
LP->HP: h_HP(n) = (-1)^n h_LP(n) ; H_HP(z)=H_LP(-z) ; flip odd powers of z^-1 ; p ->
FIR design: H(w)=b0+b1 e^{-jw}+b2 e^{-j2w}; null H(w0)=0; DC: sum b=1; pi: b0-b1+b2=1
Notch (FIR):  b0(1 - 2cos(w0) z^-1 + z^-2)
Notch (IIR):  b0(1 - 2cos w0 z^-1 + z^-2)/(1 - 2r cos w0 z^-1 + r^2 z^-2); width ~ 2(
Resonator:    b0(1 - z^-2)/(1 - 2r cos w0 z^-1 + r^2 z^-2)
Comb/MA:      (1/(M+1)) (1 - z^-(M+1))/(1 - z^-1); nulls F_k = k Fs/(M+1)
              |H| = (1/(M+1)) |sin((M+1)w/2)/sin(w/2)|, phase -Mw/2
Allpass:      z^-N A(z^-1)/A(z); poles/zeros reciprocal; 1st order (a+z^-1)/(1+a z^-1
              tau(w) = (1-a^2)/(1+2a cos w + a^2)
Oscillator:   b0/(1 - 2cos w0 z^-1 + z^-2); b0=sin w0 -> h(n)=sin(w0(n+1))u(n)
              y(n) = 2cos w0 y(n-1) - y(n-2) + b0 x(n)
INVERSE: H_I = 1/H ; poles of H_I = zeros of H
  zeros all inside  -> minimum phase    -> stable inverse causal
  zeros all outside -> maximum phase    -> stable inverse anti-causal
  mixed             -> nonminimum phase -> stable inverse non-causal (ring ROC)
  zero ON circle    -> no stable inverse
STABLE INVERSE h_I(n): partial fractions A_i/(1-p_i z^-1), A_i = [(1-p_i z^-1)H_I]_{z
  |p_i|<1 -> A_i p_i^n u(n) ;  |p_i|>1 -> -A_i p_i^n u(-n-1)
  equal degrees -> + K delta(n), K = last num coef / last den coef
  check: sum A_i + K = H_I(z->inf)
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